```
!python --version
!echo "code by: Tanvir Anjom Siddique, Lecturer, Dept. of CSE, BAUET"
```

```
Python 3.13.5
"code by: Tanvir Anjom Siddique, Lecturer, Dept. of CSE, BAUET"
```

Jupyter Notebook: K-Means Clustering Lab

What is K-Means?

K-Means is an unsupervised machine learning algorithm used to group similar data points into clusters.

- "K" means the number of clusters
- "Means" refers to the average (centroid) of each cluster

It does not require labeled data. Instead, it finds patterns based on similarity.

What K-Means Does

K-Means tries to:

1. Divide data into K groups
2. Each group has a center (called centroid)
3. Assign each data point to the nearest centroid
4. Update centroid based on assigned points

Goal: Minimize distance between points and their centroid

How K-Means Works (Concept)

Steps:

1. Choose K (number of clusters)
2. Initialize K centroids randomly
3. Assign each point to nearest centroid
4. Compute new centroids (mean of assigned points)
5. Repeat until centroids do not change

Example Data

Consider 6 points:

A(1,1), B(2,1), C(4,3), D(5,4), E(3,2), F(8,7)

Let K = 2

Initial centroids: C1 = (1,1), C2 = (5,4)

Iteration 1 (Distance Table)

Point	Dist to C1	Dist to C2	Assigned Cluster
A(1,1)	0	√25 = 5	C1
B(2,1)	1	√20 ≈ 4.47	C1
C(4,3)	√13 ≈ 3.6	√2 ≈ 1.41	C2
D(5,4)	√25 = 5	0	C2
E(3,2)	√5 ≈ 2.23	√5 ≈ 2.23	C1 (tie → choose C1)
F(8,7)	√85 ≈ 9.2	√18 ≈ 4.24	C2

New Centroids

Cluster C1: A, B, E New C1 = ((1+2+3)/3 , (1+1+2)/3) = (2, 1.33)

Cluster C2: C, D, F New C2 = ((4+5+8)/3 , (3+4+7)/3) = (5.67, 4.67)

Iteration 2

Repeat distance calculation using new centroids.

Continue until no change in assignments.

Where K-Means is Used

- Customer segmentation
- Image compression
- Document clustering
- Anomaly detection
- Market analysis

Where K-Means Cannot Be Used

- Non-spherical clusters
- Different cluster sizes
- Presence of noise/outliers
- Categorical data (without preprocessing)

Evaluation Metrics

1. Within Cluster Sum of Squares (WCSS)

$$WCSS = \Sigma \Sigma ||xi - \mu j||^2$$

Where:

- xi = data point
- μj = centroid

Lower WCSS = better clustering

2. Silhouette Score

S = (b - a) / max(a, b)

Where:

- a = average distance within cluster
- b = average distance to nearest cluster

Range: -1 to 1 Higher is better

Selecting Optimal K

Elbow Method

1. Run K-Means for different K values
2. Plot K vs WCSS
3. Find elbow point

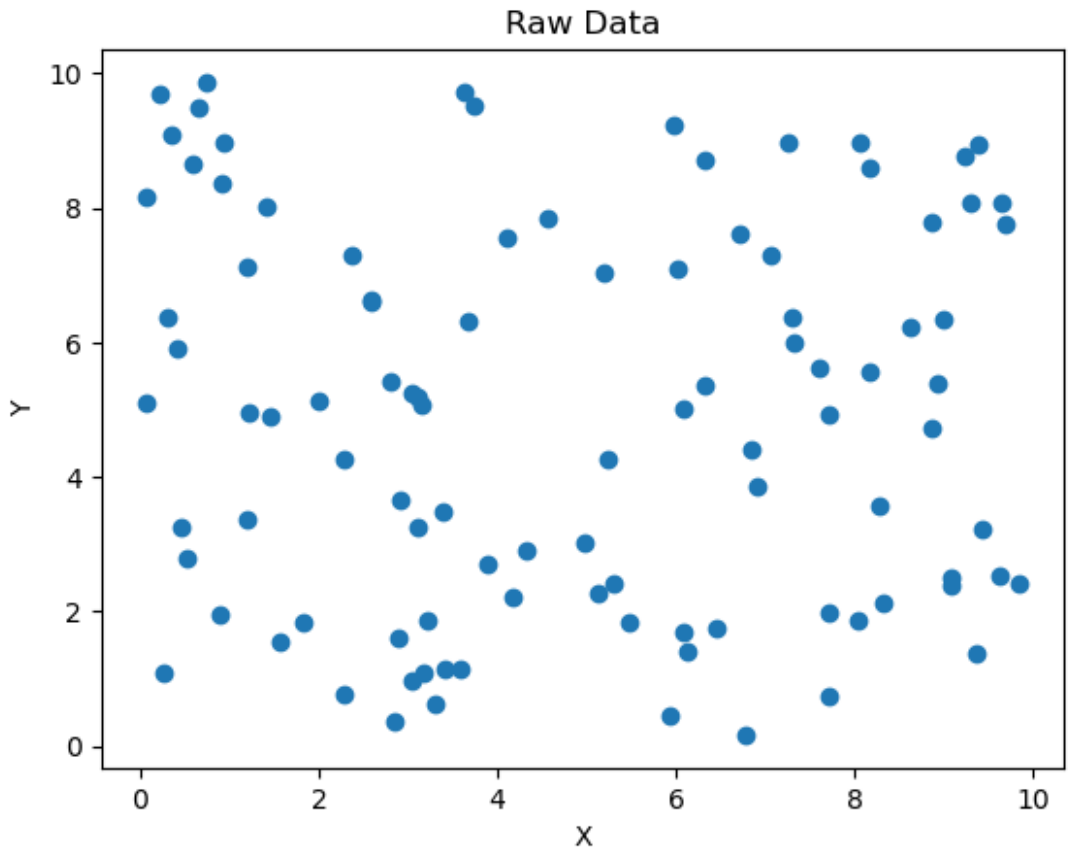
Silhouette Method

Choose K with highest silhouette score

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
data = np.random.rand(100, 2) * 10

print(data[:5])
```

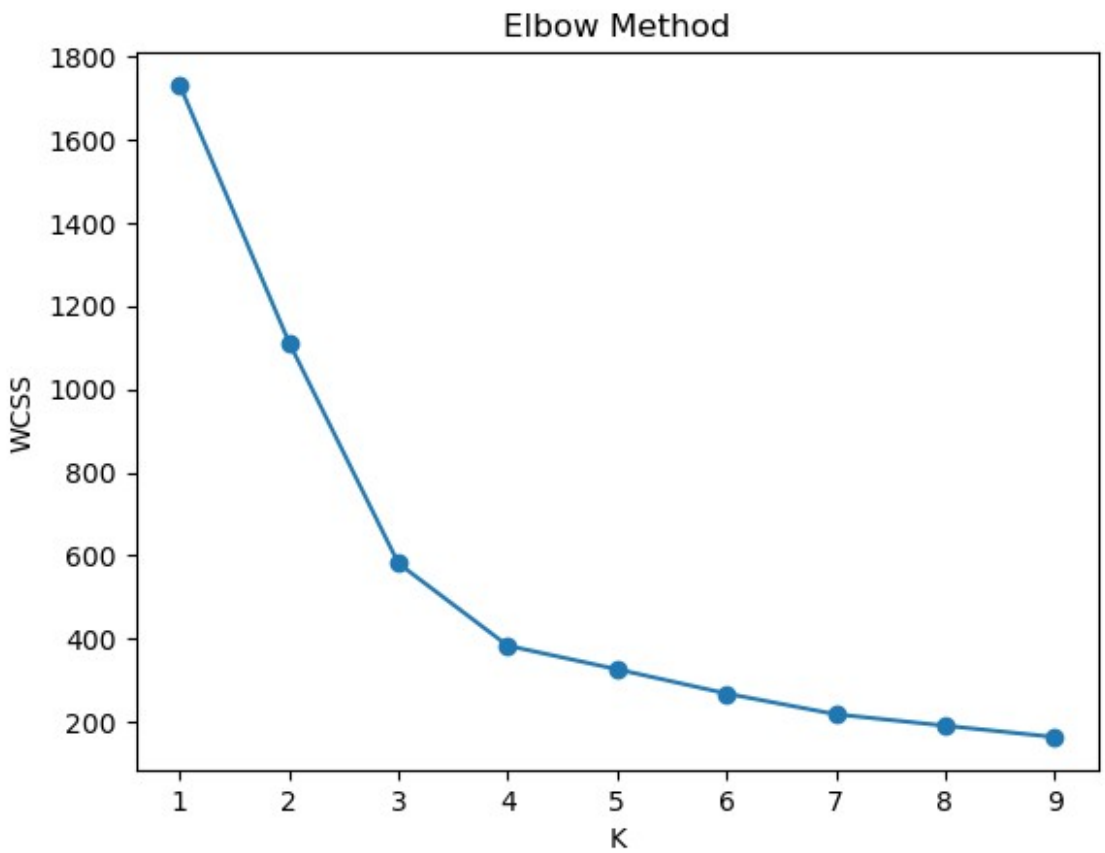
```
[[3.74540119 9.50714306]
 [7.31993942 5.98658484]
 [1.5601864  1.5599452 ]
 [0.58083612 8.66176146]
 [6.01115012 7.08072578]]
```

```
plt.scatter(data[:,0], data[:,1])
plt.title("Raw Data")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```




```
def kmeans(X, k, max_iters=100):
    centroids = X[np.random.choice(len(X), k, replace=False)]

    for _ in range(max_iters):
        distances = np.linalg.norm(X[:, None] - centroids, axis=2)
        labels = np.argmin(distances, axis=1)

        new_centroids = np.array([X[labels == i].mean(axis=0) for i in
range(k)])

        if np.all(centroids == new_centroids):
            break

        centroids = new_centroids

    return labels, centroids

def compute_wcss(X, k):
    labels, centroids = kmeans(X, k)
    wcss = 0

    for i in range(k):
        cluster_points = X[labels == i]
        wcss += np.sum((cluster_points - centroids[i])**2)
```

```
    return wcss

wcss_values = []
k_range = range(1, 10)

for k in k_range:
    wcss_values.append(compute_wcss(data, k))

plt.plot(k_range, wcss_values, marker='o')
plt.title("Elbow Method")
plt.xlabel("K")
plt.ylabel("WCSS")
plt.show()
```




```
from sklearn.metrics import silhouette_score

# already computed
k = 4
labels, centroids = kmeans(data, k)

print("Centroids:\n", centroids)
print("Labels:\n", labels)
```

```python
# --- WCSS Calculation ---
wcss = 0
for i in range(k):
    cluster_points = data[labels == i]
    wcss += np.sum((cluster_points - centroids[i])**2)

print("WCSS:", wcss)

# --- Silhouette Score ---
sil_score = silhouette_score(data, labels)
print("Silhouette Score:", sil_score)
```

```
Centroids:
 [[1.85209431 7.22280654]
 [7.8378364  7.3812559 ]
 [7.51187305 2.57045258]
 [2.74520439 2.1308078 ]]
Labels:
 [0 1 3 0 1 0 2 3 0 3 2 3 0 0 2 2 0 1 3 2 0 0 0 0 2 1 1 1 3 3 3 2 0 0
0 2 0
 1 2 3 1 3 3 1 2 0 1 2 2 3 0 0 2 0 3 3 1 1 2 1 1 3 3 1 0 3 3 2 1 0 2 3
3 2
 3 2 0 2 1 0 0 1 0 3 0 2 3 2 2 2 3 1 0 1 2 0 1 3 1 1]
WCSS: 378.18689301285934
Silhouette Score: 0.42430208228214
```

Explanation (for students)

- WCSS measures how close points are to their centroid
- Lower value means better clustering
- Silhouette Score measures how well-separated clusters are
- Value close to 1 means good clustering

```python
# Classifying a New Unknown Point
# new unseen data point
new_point = np.array([6, 5])

# compute distance from each centroid
distances = np.linalg.norm(centroids - new_point, axis=1)

# find nearest centroid
assigned_cluster = np.argmin(distances)

print("New Point:", new_point)
print("Distances from centroids:", distances)
print("Assigned Cluster:", assigned_cluster)
```

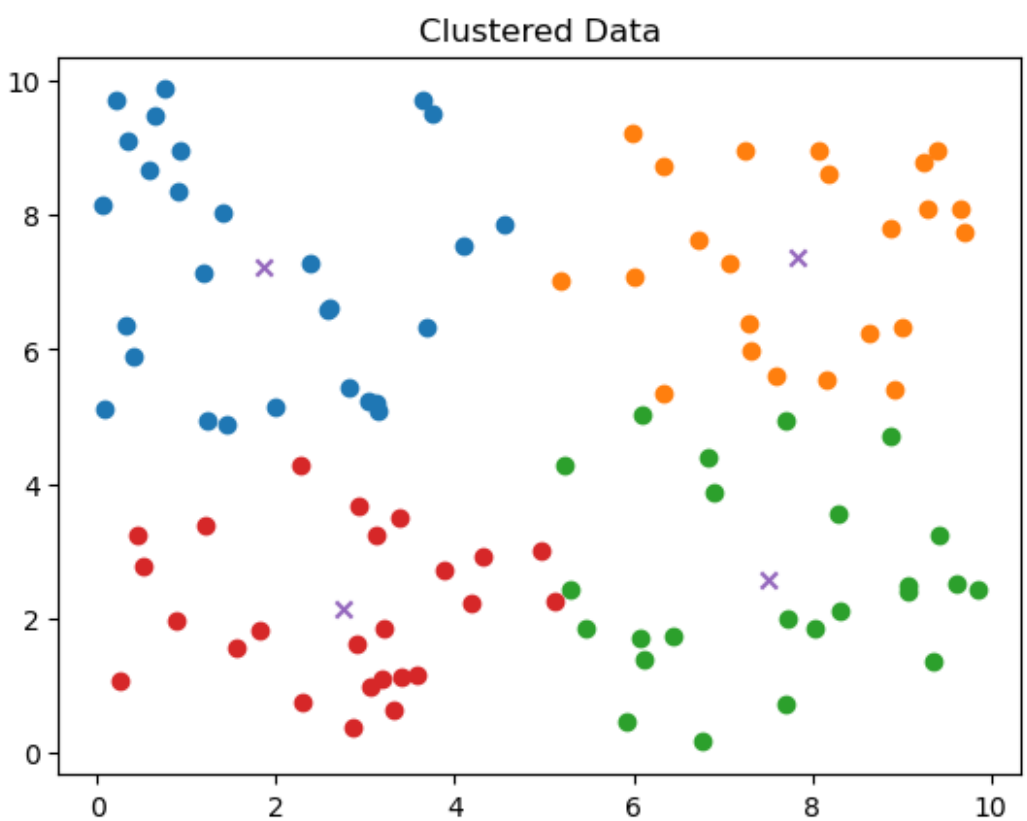
```
New Point: [6 5]
Distances from centroids: [4.70595267 3.00799307 2.86154867
```

```
4.33888907]
Assigned Cluster: 2
```

Explanation:

- The new point is compared with all centroids
- Distance is calculated
- The nearest centroid determines its cluster

```
for i in range(k):
    plt.scatter(data[labels == i][:,0], data[labels == i][:,1])

plt.scatter(centroids[:,0], centroids[:,1], marker='x')
plt.title("Clustered Data")
plt.show()
```

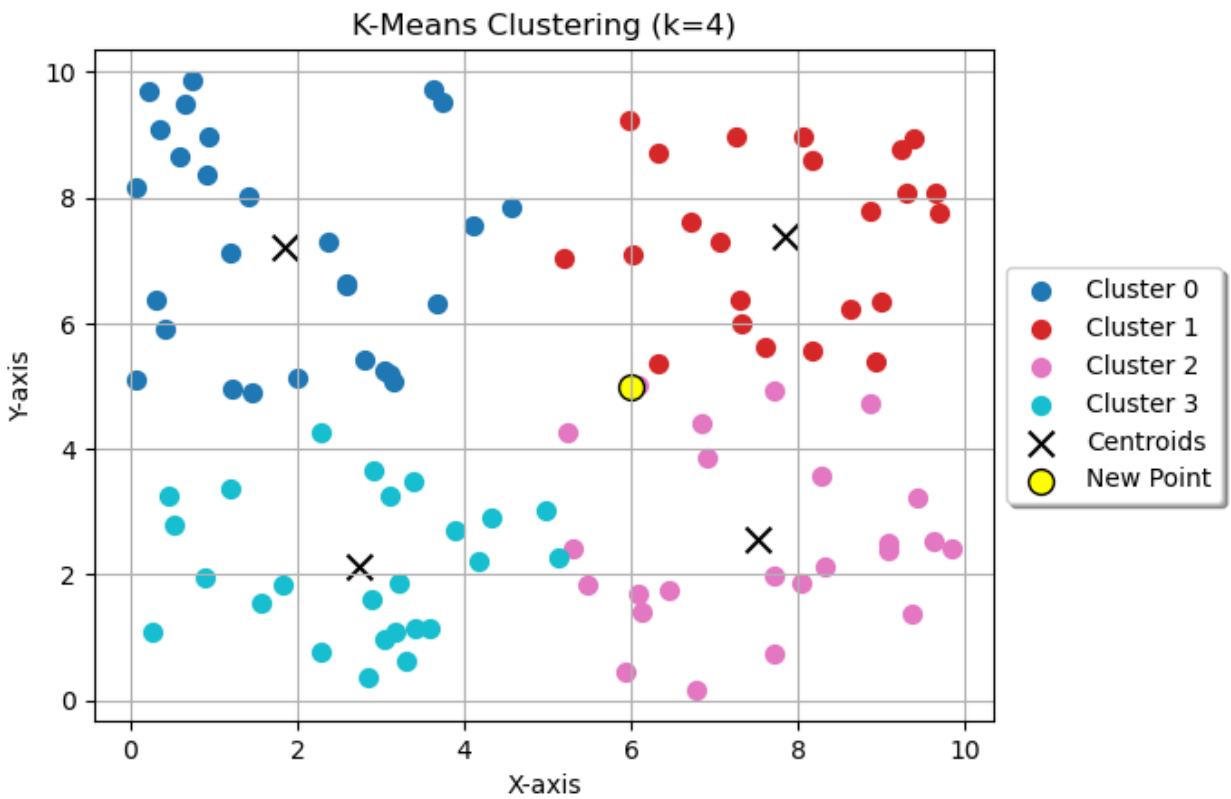


Plot existing and new points

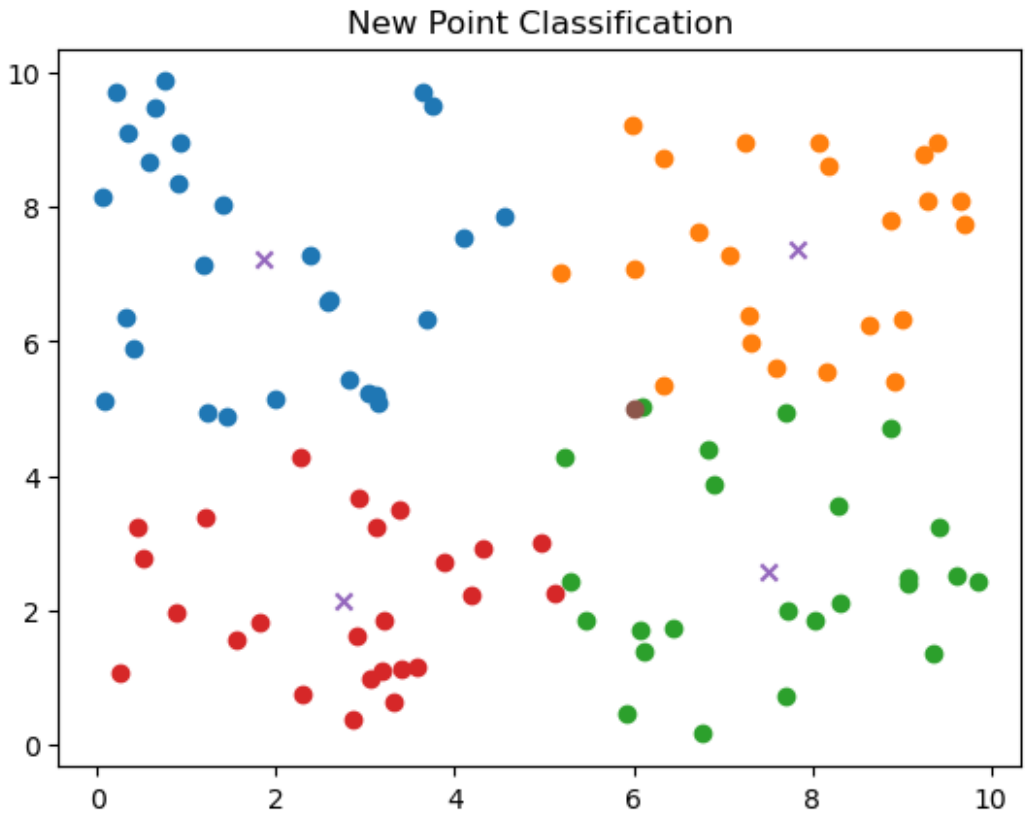
```
# ensure inline plotting in Jupyter
# makes your graphs appear directly inside the notebook cell, instead
of opening in a separate window
%matplotlib inline
import matplotlib.pyplot as plt
import matplotlib.cm as cm
```

```python
import numpy as np

# Example: k, labels, centroids, new_point already computed
# k = 4
# labels, centroids = kmeans(data, k)
# new_point = np.array([6, 5])

# choose colormap for k clusters
cmap = cm.get_cmap('tab10', k)

# plot each cluster dynamically
for i in range(k):
    plt.scatter(
        data[labels == i][:, 0],
        data[labels == i][:, 1],
        color=cmap(i),
        label=f'Cluster {i}',
        s=50
    )

# plot centroids
plt.scatter(
    centroids[:, 0],
    centroids[:, 1],
    marker='x',
    s=100,
    color='black',
    label='Centroids'
)

# plot new unseen point
plt.scatter(
    new_point[0],
    new_point[1],
    marker='o',
    s=100,
    color='yellow',
    edgecolor='black',
    label='New Point'
)

# labels, title, legend
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title(f"K-Means Clustering (k={k})")
# plt.legend()
# Legend on the right, outside the plot
plt.legend(
    loc='center left',          # anchor point on plot's left center
    bbox_to_anchor=(1, 0.5),    # move legend outside to right
```

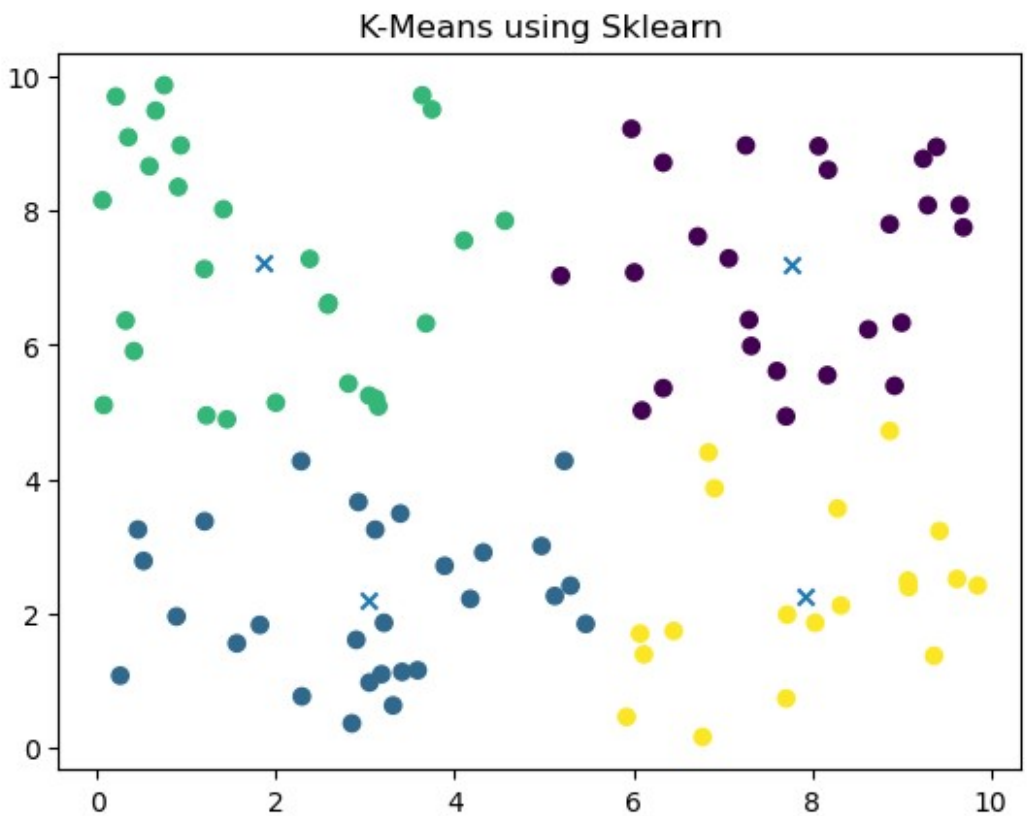
```
    ncol=1,                         # 1 column
    fancybox=True,                  # rounded legend box
    shadow=True
)
plt.grid(True)

# show the plot
plt.show()
```




```
# plot existing clusters
for i in range(k):
    plt.scatter(data[labels == i][:,0], data[labels == i][:,1])

# plot centroids
plt.scatter(centroids[:,0], centroids[:,1], marker='x')

# plot new point
plt.scatter(new_point[0], new_point[1], marker='o')

plt.title("New Point Classification")
plt.show()
```




```python
import warnings
warnings.filterwarnings("ignore")

import os
os.environ["OMP_NUM_THREADS"] = "1"
# UserWarning: KMeans is known to have a memory leak on Windows with
MKL, when there are less chunks than available threads.
# You can avoid it by setting the environment variable
OMP_NUM_THREADS=1.
from sklearn.cluster import KMeans

kmeans_model = KMeans(n_clusters=4, random_state=42)
kmeans_model.fit(data)

labels = kmeans_model.labels_
centroids = kmeans_model.cluster_centers_

print("Centroids:\n", centroids)
```

```
Centroids:
 [[7.76302209 7.18934527]
 [3.02218023 2.20759545]
 [1.85209431 7.22280654]
 [7.92045796 2.27282306]]
```

```
plt.scatter(data[:,0], data[:,1], c=labels)
plt.scatter(centroids[:,0], centroids[:,1], marker='x')
plt.title("K-Means using Sklearn")
plt.show()
```